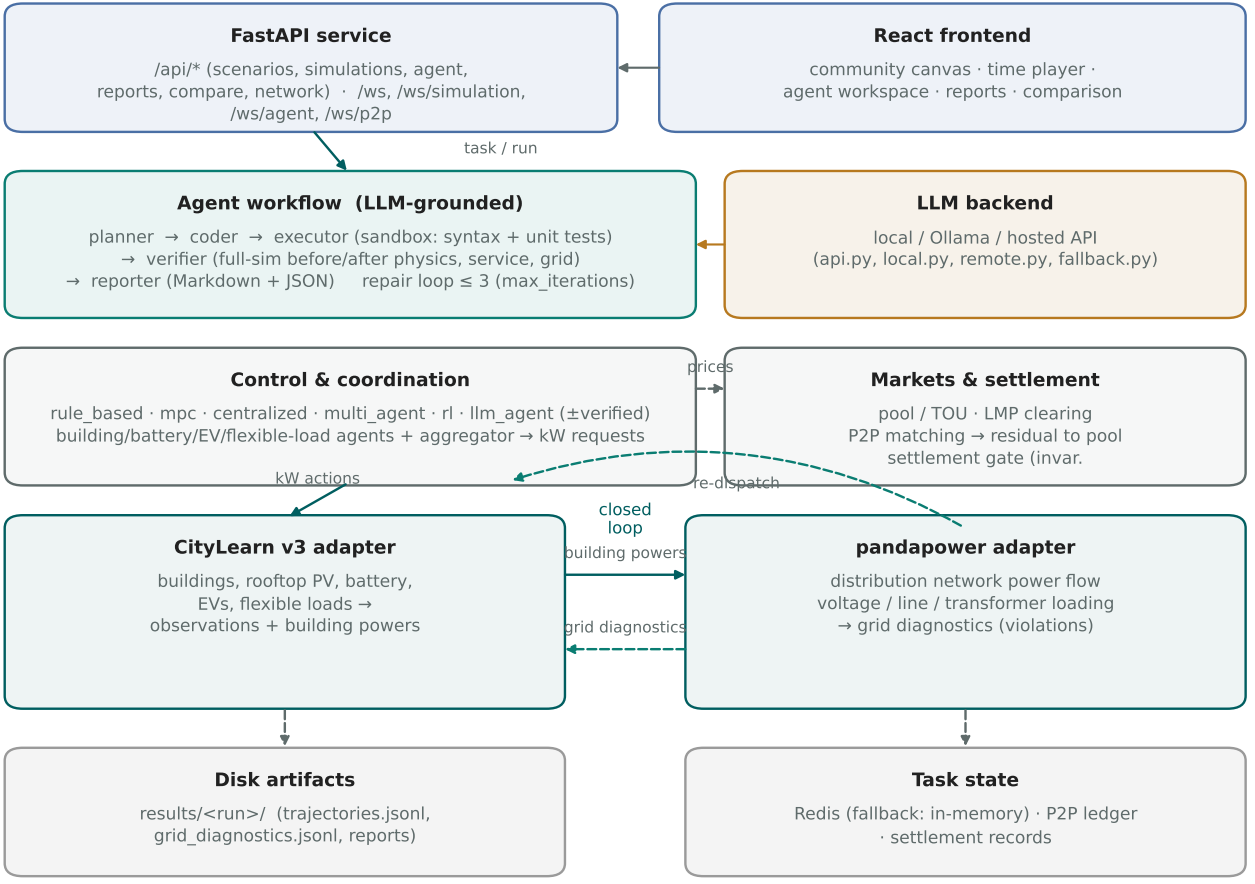
FastAPI service
/api/* (scenarios, simulations, agent, reports, compare, network) · /ws, /ws/simulation, /ws/agent, /ws/p2p
React frontend
community canvas · time player · agent workspace · reports · comparison
task / run
Agent workflow (LLM-grounded)
planner → coder → executor (sandbox: syntax + unit tests) → verifier (full-sim before/after physics, service, grid) → reporter (Markdown + JSON) repair loop ≤ 3 (max_iterations)
LLM backend
local / Ollama / hosted API (api.py, local.py, remote.py, fallback.py)
Control & coordination
rule_based · mpc · centralized · multi_agent · rl · llm_agent (±verified) building/battery/EV/flexible-load agents + aggregator → kW requests
prices
Markets & settlement
pool / TOU · LMP clearing P2P matching → residual to pool settlement gate (invar.
kW actions
re-dispatch
closed loop
CityLearn v3 adapter
buildings, rooftop PV, battery, EVs, flexible loads → observations + building powers
building powers
grid diagnostics
pandapower adapter
distribution network power flow voltage / line / transformer loading → grid diagnostics (violations)
Disk artifacts
results/<run>/ (trajectories.jsonl, grid_diagnostics.jsonl, reports)
Task state
Redis (fallback: in-memory) · P2P ledger · settlement records